SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

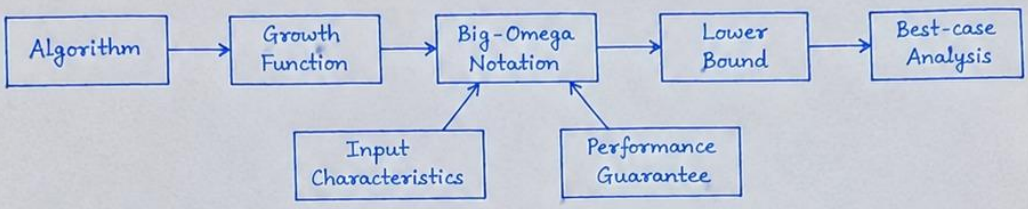
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

7. Consider the concept map: Algorithm → Growth Function → Big-Omega Notation → Lower Bound → Best-case Analysis. Explain the relationships among the nodes and show how lower-bound analysis captures the minimum behavior of an algorithm. Extend the map by linking Performance Guarantee and Input Characteristics as additional supporting nodes. Justify why Big-Omega is useful when discussing guaranteed minimum efficiency. Interpret how the concept map gives a balanced understanding of algorithm behavior from the lower-bound perspective.

Answer:

The completed concept map is shown below:



1. Algorithm → Growth Function :
   - The algorithm's basic operations are analyzed to form a growth function (e.g., $f(n)$).
   - This function shows how the running time increases with the input size.
2. Growth Function → Big-Omega Notation :
   - The growth function is expressed using Big-Omega (Ω) notation.
   - Big-Omega represents a lower bound on the running time.
3. Big-Omega Notation → Lower Bound :
   - The Ω notation gives the minimum rate of growth.
   - It ensures that the algorithm takes at least this much time for large inputs.
4. Lower Bound → Best-case Analysis :
   - The lower bound corresponds to the best possible performance (minimum time).
   - Best-case analysis shows the scenario where the algorithm performs the fastest.
5. Supporting Nodes :
   - Input Characteristics: The nature and size of the input can influence the growth function and hence the lower bound.
   - Performance Guarantee: It provides assurance that the algorithm will not perform worse than the bound given by Big-Omega.

Justification:

- Big-Omega is useful because it gives a guaranteed minimum efficiency.
- It helps in setting a lower limit on the running time, which is important for real-world applications where performance must not fall below a certain level.
- Without Big-Omega, we cannot ensure that the algorithm performs at least as well as expected.

Interpretation:

This concept map provides a balanced view of algorithm behavior by showing both the theoretical lower bound (Big-Omega), and the practical aspects (input characteristics and performance guarantee), along with the best-case scenario. It helps in understanding not just the average or worst performance, but also the minimum guaranteed efficiency of the algorithm.